

Accessing dynamic functional connectivity using ℓ_0 -regularized sparse-smooth inverse covariance estimation from fMRI

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ABSTRACT

Inferring dynamic functional connectivity (dFC) from functional magnetic resonance imaging (fMRI) is crucial to understand the time-variant functional inter-relationships among brain regions. Because of the sparse property of functional connectivity networks, sparsity-promoting dFC estimation methods, which are mainly based on ℓ_1 -norm regularization, are gaining popularity. However, ℓ_1 -norm regularization cannot provide the maximum sparsity solution as the most natural sparsity promoting norm, the ℓ_0 -norm. But ℓ_0 -norm is seldom used to infer sparse dFC because an efficient algorithm to address the non-convexity problem of ℓ_0 -norm is lacking. In this work, we develop a new ℓ_0 -norm regularization-based inverse covariance estimation method for estimating dFC from fMRI. This novel method employs ℓ_0 -norm regularizations on both spatial and temporal scales to enhance the spatial sparsity and temporal smoothness of dFC estimates. To overcome the non-convexity of ℓ_0 -norm, we further propose an effective optimization algorithm based on the coordinate descent (CD). The performance of the proposed ℓ_0 -norm-based sparse-smooth regularization (L0-SSR) method is examined using a series of synthetic datasets concerning various types of network topology. We further apply the proposed L0-SSR method to real fMRI data recorded in block-design motor tasks from 45 participants for the exploration of task induced dFC. Results on synthetic and real-world fMRI data show that, the L0-SSR method can achieve more accurate and interpretable dFC estimates than conventional ℓ_1 -norm-based dFC estimation methods. Hence, the proposed L0-SSR method could serve as a powerful analytical tool to infer highly complex, variable, and sparse dFC patterns.

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1. Introduction

The human brain works as a complex and high-efficiency network with numerous inter-connected brain regions. To understand the underlying functional mechanism of the brain network and the inter-relationships between brain regions, the study of functional connectivity (FC) has gained popularity. Generally, FC can be regarded as the statistical interdependence between two brain

regions [28], and functional connectivity networks (FCNs) can be defined by the presence, absence, and strength of FC among multiple brain regions. FC is commonly estimated using blood oxygenation level dependent (BOLD) signals from functional magnetic resonance imaging (fMRI) [28,53]. An important advantage of fMRI in estimating FC is that it enables the observations of brain activity localized in both superficial and deep sources with relatively high spatial resolution [54]. Conventionally, FC and FCN are assumed to be static during the whole scanning period. Based on this assumption, various FC estimation methods, including seed-based approach, multivariate regressive model, independent component analysis, and principle component analysis, have been proposed [29,12,16,62].

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To date, it is widely understood that the FC is dynamic rather than static, even during the resting-state [14,3,37]. Many approaches have been developed to estimate dynamic FC (dFC), and among them the most popular one is the sliding window-based estimation method [3,46]. The sliding window-based method partitions the entire fMRI time-series into multiple temporal segments. By calculating the inter-dependence (normally using Pearson correlation coefficient, covariance, inverse covariance, etc.) between two regions of interest (ROIs) for each segment, one can obtain dFC estimates for the entire time-series [3,51,7,40]. Besides the sliding window-based method, some other approaches for dFC estimation have been developed. For example, hidden Markov model-based method is an alternative approach for estimating dynamic FCNs, which assume the brain activity has several discrete states characterized by different connectivity patterns [63,4]. Another popular framework is dynamic connectivity regression, which estimate dynamic FCNs in two steps: it first detects the changing points and then estimates a graph in the temporal partition that falls between pairs of changing points [19,20]. In recent years, a lot of attention has been drawn to multivariate statistical methods for dFC inferring problem [42,41,15]. The advantage of such methods is that it can estimate the FCN at each timepoint for the entire time series so that more details can be characterized over time.

One important property of the FCNs is spatial sparse, which means only a few brain regions are inter-connected during a certain condition so that the brain can achieve a high efficiency of information transfer [51,14,33,31,34]. If we consider the FCN as a graph, where ROIs are nodes and FCs are edges, the sparsity of an FCN can be referred to the sparsity of the adjacency matrix describing it. While introducing the sparsity, relevant and meaningful connectivity and associated regions can be better preserved and estimated since those spurious connections are set to 0 and thus are ignored during the estimation [25]. Aside from the spatial sparsity, another essential property of dFC is temporal smoothness, which means the FCN would not sharply reconfigure but should maintain its structure within a relatively short period [42,35,41]. More specifically, temporal smoothness indicates that the temporal adjacent FCNs would smoothly changes with high similarity, which also meet the assumption of sliding window approach. In another word, the difference among FC matrices of neighboring timepoints should be sparse, i.e. most entries of the difference matrix should be zero.

Due to its advantages of convexity and simplicity, ℓ_1 -norm-based regularization has been widely used to introduce spatial sparsity [3,1] and temporal smoothness in dFC estimation [42;35,15]. The ℓ_1 -norm enjoys attractive statistical properties under certain regularity conditions since it produces model with good prediction accuracy when the underlying model is reasonable sparse. However, these regularities are easily violated since ℓ_1 -norm does both shrinkage and selection at the same time. It often includes more non-vital coefficients thus selects an overly dense model with large bias due to the relaxation on the relevant coefficients, which could lead to false positive [40,48,26,64]. This may become a significant issue in the estimation of dynamic FCN. It is because FCNs are typical large-scale networks, and a small discovery rate of false positive will largely affect the estimation results [42]. In these cases, greedier methods like non-convex approaches achieve sparser models than ℓ_1 -norm for the same or better performance and enjoy superior variable-selection properties [39]. To overcome the limitations of ℓ_1 -norm, many researchers in the signal processing field have proposed to use ℓ_0 -norm because ℓ_0 -norm can introduce maximum sparsity without shrinking and has superior performance compared with ℓ_1 -norm especially for very sparse problem [52,30,57,18]. ℓ_0 -norm has already been successfully used in a wide variety of applications, such as compressed

sensing [22,57] and neural network learning [36]. It is natural to come out a question that whether ℓ_0 -norm can be applied to the dFC estimation and provides improved performance than conventional ℓ_1 -norm by introducing better spatial sparsity and temporal smoothness.

In this work, we develop a novel dFC estimation method called ℓ_0 -norm-based sparse-smooth regularization (LO-SSR), which can overcome the limitations of conventional ℓ_1 -norm-based dFC estimation methods. First, to achieve the spatial sparsity property of FCN, we propose to use an ℓ_0 -norm-based spatially sparse regularization on the entire FC matrix at each time point to obtain significant FC while eliminate noisy estimates. Further, we employ an ℓ_0 -norm-based temporally smooth regularization on the difference matrix between temporally-consecutive adjacency matrices to control the temporal variability of FCN (i.e., to keep remarkable FC changes while ignore small perturbations). Considering the non-convexity of the ℓ_0 -norm regularization terms, a novel and efficient algorithm based on coordinate descent (CD) [38] is proposed for LO-SSR so that the estimator can be solved by coordinate-by-coordinate optimization. By utilizing this CD algorithm, one can find the local minimizer of the objective function of LO-SSR iteratively with high efficiency.

The performance of the proposed LO-SSR method on is firstly examined using synthetic data, which employed various graph sizes and models to simulate different types of real-world FCN. Next, the LO-SSR method is applied to a motor-task fMRI dataset, in which fMRI data of 45 healthy subjects were recorded when executing a series of block-design tasks to engage motor actions involving fingers, toes and tongue. Results on both synthetic and real-world fMRI data show that, the proposed LO-SSR method can achieve superior results than conventional ℓ_1 -norm-based methods.

The rest of the paper is organized as follows. In Section 2, the proposed LO-SSR method and the CD-based algorithm are introduced. Section 3 is devoted to the simulation studies of the proposed method. The performance of the proposed method on real-world fMRI data is provided in Section 4. Finally, discussions and conclusions are presented in Section 5.

2. Methods

In this section, we will present the formulation of the LO-SSR dFC estimation method and the corresponding CD optimization algorithm. In the following content, we will use the notations as described in Table 1.

2.1. Mathematical model

Suppose we have n -dimensional BOLD signals at T timepoints. We assume that the BOLD signals at each timepoint follow a mul-

Table 1
Frequently used notations in this paper.

$\mathbf{X}_t$	A square matrix at timepoint t .
$[\mathbf{X}_t]_{ij}$ or $x_{ij,t}$	The i -th row and j -th column entry from a matrix $\mathbf{X}_t$.
$\mathbf{X}^T$	The transpose of a matrix $\mathbf{X}$.
$\det(\mathbf{X})$	The determinant of a matrix $\mathbf{X}$.
$\text{tr}(\mathbf{X})$	The trace of a matrix $\mathbf{X}$.
$\text{diag}(\mathbf{X})$	The vector with all diagonal entries of $\mathbf{X}$.
$\mathbf{X} \succ 0$ and $\mathbf{X} \succeq 0$	The matrix $\mathbf{X}$ is positive definite and positive semi-definite respectively.
$\mathbf{e}_i$	A column vector with the i -th entry as 1 and all others as 0.
$\mathbf{U}_{ij}$	A matrix with two columns that are defined as $[\mathbf{e}_i, \mathbf{e}_j]$.
$\mathbb{I}\{\cdot\}$	The logical indicator function, returns 1 if the statement in the brackets is true and 0 otherwise.
$\ \mathbf{X}\ _0$	The ℓ_0 -norm of $\mathbf{X}$, $\ \mathbf{X}\ _0 = \sum_i \sum_j \mathbb{I}\{x_{ij} \neq 0\}$.

tivariate Gaussian distribution with mean μ and covariance $\Sigma_{n \times n}$. Let the time-varying sample covariance matrix at timepoint t be denoted as $\mathbf{S}_t$ and the time-varying inverse covariance matrix, a.k.a. precision matrix, at timepoint t be denoted as $\mathbf{X}_t = \Sigma_t^{-1} \succ 0, t = 1, \dots, T$. Additionally, we assume that the diagonal elements of $\mathbf{S}_t$ are all positive. By using the maximum likelihood estimation, we aim to find the optimal $\mathbf{X}_t$, which is symmetric and positive definite, at each timepoint to minimize the following log-likelihood problem

$$\mathcal{L} = -\log \det \mathbf{X}_t + \text{tr}(\mathbf{S}_t \mathbf{X}_t). \quad (1)$$

Conventionally, the time-varying sample covariance matrices can be obtained from truncated data by the sliding window-based method. The above expression in (1) is called the Gaussian graphical model and the estimated $\mathbf{X}_t$ can be used as the adjacency matrix to represent a graph where the off-diagonal entries $[\mathbf{X}_t]_{ij}$ can be regarded as the edges between node i and node j .

From the mathematical point of view, spatial sparsity and temporal smoothness of dFC networks mean that the adjacency matrix at each timepoint should be sparse, and the difference between two neighboring adjacency matrices should be sparse as well. To achieve these two types of sparsity, researchers tended to add sparsity-promoting regularizations terms, such as ℓ_1 -norm, into the original model of (1) to impose constraints. In this study, to achieve the maximum sparsity, ℓ_0 -norm is incorporated into the original model in (1) for dFC estimation. The resultant objective function should read

$$\mathcal{L}_{\ell_0} = -\log \det \mathbf{X}_t + \text{tr}(\mathbf{S}_t \mathbf{X}_t) + \lambda_1 \|\mathbf{X}_t\|_0 + \lambda_2 \|\mathbf{X}_t - \mathbf{X}_{t-1}\|_0, \quad (2)$$

where $\mathbf{X}_{t-1}$ is the estimate at timepoint $t-1$, λ_1 and λ_2 are the non-negative regularization parameters to control the imposing weight for spatial sparsity and temporal smoothness of the estimated coefficients, respectively. Since the proposed model of (2) imposes both sparse and smooth regularizations using ℓ_0 -norm, it is referred to as the ℓ_0 -norm-based sparse-smooth regularization, or the L0-SSR method for short. It should be noted that the two consecutive regularization terms have different purposes. The first regularization, $\lambda_1 \|\mathbf{X}_t\|_0$, is used to enforce the spatial sparsity of the FCN itself at timepoint t . On the other hand, the second regularization, $\lambda_2 \|\mathbf{X}_t - \mathbf{X}_{t-1}\|_0$, handles the difference between two temporal neighboring brain networks to enforce them to be similar.

2.2. CD algorithm

Due to the non-linear and non-convex properties of the model in (2), the optimal solution for (2) cannot be easily achieved with conventional algorithms, such as traditional Gauss-Newton algorithm. To address this, we develop a new iterative algorithm based on coordinate descent (CD), which minimizes only one entry with all others fixed in each iteration. Once the selected minimizer is found, the objective coefficient matrix $\mathbf{X}_t$ will be updated and continue to the next selected entry in the following iteration. Such an update procedure can be represented as

$$\mathbf{Z}(x_{ij}^{k+1}) = \mathbf{X}^k + \begin{cases} \delta(x_{ii}^{k+1}) \mathbf{e}_i \mathbf{e}_i^T & \text{if } i = j \\ \delta(x_{ij}^{k+1}) \mathbf{u}_{ij} \mathbf{u}_{ij}^T & \text{otherwise,} \end{cases} \quad (3)$$

where $\mathbf{Z}(x_{ij}^{k+1})$ denotes the updated matrix with minimized entry x_{ij} , function $\delta(\cdot)$ is defined as $\delta(x_{ij}^{k+1}) = x_{ij}^{k+1} - x_{ij}^k$, the superscript k denotes the iteration number. For notational simplicity, the time-point indicator t is omitted in (3) and hereinafter unless otherwise

specified. Thus, for any pair of indices i and j , the objective function to be minimized can be generally expressed as

$$\Psi(x_{ij}^{k+1}) = -\log \det(\mathbf{Z}(x_{ij}^{k+1})) + d_{ij} s_{ij} x_{ij}^{k+1} + d_{ij} \lambda_1 \mathbb{I}\{x_{ij}^k \neq 0\} + d_{ij} \lambda_2 \mathbb{I}\{x_{ij}^k \neq x_{ij,t-1}\}, \quad (4)$$

where s_{ij} is the ij -th element of the sample covariance matrix $\mathbf{S}$, d_{ij} is equal to 1 if $i = j$ and equal to 2 if $i \neq j$. In the following, we will discuss the initialization and minimization procedures under two circumstances: $i = j$ and $i \neq j$.

2.2.1. Initialization

It is suggested in [38] that the method should be initialized with a very sparse matrix, e.g. a diagonal matrix. In our work, we follow the method proposed in [38] as the initialization procedure. At the first timepoint, we calculate $\mathbf{X}_1$ using the static inverse covariance estimation method. At the following timepoints from $t = 2, \dots, T$, we initialize $\mathbf{X}_t$ as $\text{diag}(1/s_{11,t}, 1/s_{22,t}, \dots, 1/s_{nn,t})$.

2.2.2. Minimization of Ψ when $i = j$

If $i = j$, the selected entries will be lay on the diagonal of the coefficient matrices. Under such a circumstance, the resultant optimization problem becomes

$$\arg \min_{x_{ii}^{k+1}} \Psi(x_{ii}^{k+1}) = \arg \min_{x_{ii}^{k+1}} -\log \det(\mathbf{Z}(x_{ii}^{k+1})) + s_{ii} x_{ii}^{k+1} + \lambda_1 + \lambda_2 \mathbb{I}\{x_{ii}^{k+1} \neq x_{ii,t-1}\}. \quad (5)$$

If $x_{ii,t-1}$ is in the domain of the objective function, i.e. $\det(\mathbf{Z}(x_{ii}^{k+1})) > 0$, the function has one discontinuous point at $x_{ii}^{k+1} = x_{ii,t-1}$ since the last term $\lambda_2 \mathbb{I}\{x_{ii}^{k+1} \neq x_{ii,t-1}\}$ is equal to zero, which will lead to a sudden change. Meanwhile, $\Psi(\cdot)$ will be continuous if $x_{ii,t-1}$ is not in the domain of the function.

When $x_{ii}^{k+1} \neq x_{ii,t-1}$, we define the continuous part of $\Psi(\cdot)$ as

$$c(x_{ii}^{k+1}) = -\log \det(\mathbf{Z}(x_{ii}^{k+1})) + s_{ii} x_{ii}^{k+1} + \lambda_1 + \lambda_2. \quad (6)$$

Firstly, if $x_{ii,t-1}$ is in the domain of the objective function, i.e. $\det(\mathbf{Z}(x_{ii}^{k+1})) > 0$, the objective function in (5) can be expressed by using (6) as

$$\Psi(x_{ii}^{k+1}) = c(x_{ii}^{k+1}) \mathbb{I}\{x_{ii}^{k+1} \neq x_{ii,t-1}\} + (c(x_{ii}^{k+1}) - \lambda_2) \mathbb{I}\{x_{ii}^{k+1} = x_{ii,t-1}\}. \quad (7)$$

Now we can see that the minimizers of $\Psi(\cdot)$ are the minimizers of function $c(\cdot)$ or simply $x_{ii}^{k+1} = x_{ii,t-1}$. Since (6) is convex and differentiable, the solution can be obtained by making its derivative equal to zero:

$$c'(x_{ii}^{k+1}) = -\left[\mathbf{Z}(x_{ii}^{k+1})^{-1}\right]_{ii} + s_{ii} = 0. \quad (8)$$

By making use of the Sherman-Morrison-Woodbury formula, we have

$$(\mathbf{X} + \theta \mathbf{e}_i \mathbf{e}_i^T)^{-1} = \mathbf{Y} - \frac{\theta \mathbf{Y} \mathbf{e}_i \mathbf{e}_i^T \mathbf{Y}}{1 + \theta y_{ii}} = \mathbf{Y} - \frac{\theta \mathbf{y}_{ij} \mathbf{y}_{ij}^T}{1 + \theta y_{ii}}, \quad (9)$$

where $\mathbf{e}_i$ is a vector defined in Table 1 with its i -th entry equal to 1 and all others 0, θ is a constant, $\mathbf{Y} = \mathbf{X}^{-1}$, y_{ij} stands for the ij -th element of $\mathbf{Y}$, $\mathbf{y}_{ij}$ is the i -th column of $\mathbf{Y}$. Let $\theta = \delta(x_{ii}^{k+1}) = x_{ii}^{k+1} - x_{ii}^k$, we can have

$$\left[\mathbf{Z}(x_{ii}^{k+1})^{-1}\right]_{ii} = \frac{y_{ii}^k}{1 + \delta(x_{ii}^{k+1}) y_{ii}^k}, \quad (10)$$

Consequently, by substituting (10) into (8) and solving for x , the minimizer will be given by

$$\hat{x}_{ii}^{k+1} = m_{ii}^k = x_{ii}^k + \frac{y_{ii}^k - s_{ii}}{y_{ii}^k s_{ii}}. \quad (11)$$

Secondly, if $\det(\mathbf{Z}(x_{ii}^{k+1})) \leq 0$, then $x_{ii,t-1}$ is not in the domain of the objective function. Thus, the minimizer of $\Psi(\cdot)$ is given by m_{ii}^k as in (11).

The above discussions are summarized as following:

- when $\det(\mathbf{Z}(x_{ii}^{k+1})) \leq 0$:

$$\hat{x}_{ii}^{k+1} = m_{ii}^k; \quad (12)$$

- when $\det(\mathbf{Z}(x_{ii}^{k+1})) > 0$:

$$\hat{x}_{ii}^{k+1} = \begin{cases} x_{ii,t-1} & \text{if } \Psi(x_{ii,t-1}) < \Psi(m_{ii}^k) \\ m_{ii}^k \parallel \{\hat{x}_{ii}^k \neq x_{ii,t-1}\} + x_{ii,t-1} \parallel \{\hat{x}_{ii}^k = x_{ii,t-1}\} & \text{if } \Psi(x_{ii,t-1}) = \Psi(m_{ii}^k) \\ m_{ii}^k & \text{if } \Psi(x_{ii,t-1}) > \Psi(m_{ii}^k) \end{cases}. \quad (13)$$

2.2.3. Minimization of Ψ when $i \neq j$

Next, we shall discuss a more complicated case when $i \neq j$, where the selected entries are off-diagonal. Similar to (5), we can have the minimizer of the objective function

$$\arg \min_{x_{ij}^{k+1}} \Psi(x_{ij}^{k+1}) = \arg \min_{x_{ij}^{k+1}} -\log \det(\mathbf{Z}(x_{ij}^{k+1})) + 2s_{ij}x_{ij}^{k+1} + 2\lambda_1 \parallel \{x_{ij}^{k+1} \neq 0\} + 2\lambda_2 \parallel \{x_{ii}^{k+1} \neq x_{ii,t-1}\}. \quad (14)$$

In this case, the L0-SSR method has two discontinuous points at $x_{ij}^{k+1} = 0$ and $x_{ij}^{k+1} = x_{ij,t-1}$ if they are in the domain of the objective function.

As in Section 2.2.2, we first consider the continuous case of $\Psi(\cdot)$ as

$$c(x_{ij}^{k+1}) = -\log \det(\mathbf{Z}(x_{ij}^{k+1})) + 2s_{ij}x_{ij}^{k+1} + 2\lambda_1 + 2\lambda_2. \quad (15)$$

By making the derivative equal to zero, we have

$$c'(x_{ij}^{k+1}) = -2 \left[\mathbf{Z}(x_{ij}^{k+1})^{-1} \right]_{ii} + 2s_{ij} = 0. \quad (16)$$

Again, by making use of the Sherman-Morrison-Woodbury formula, we have

$$\begin{aligned} (\mathbf{X} + \theta \mathbf{U}_{ij} \mathbf{U}_{ij}^T)^{-1} &= \mathbf{Y} - \theta \mathbf{Y} \mathbf{U}_{ij} (\mathbf{I} + \theta \mathbf{U}_{ij}^T \mathbf{Y} \mathbf{U}_{ij})^{-1} \mathbf{U}_{ij}^T \mathbf{Y} \\ &= \mathbf{Y} - \frac{\theta \begin{bmatrix} \mathbf{y}_{[i]} & \mathbf{y}_{[j]} \end{bmatrix} \begin{bmatrix} 1 + \theta y_{ij} & -\theta y_{ij} \\ -\theta y_{ij} & 1 + \theta y_{ij} \end{bmatrix} \begin{bmatrix} \mathbf{y}_{[j]}^T \\ \mathbf{y}_{[i]}^T \end{bmatrix}}{-\Delta_{ij} \theta^2 + 2y_{ij} \theta + 1}, \end{aligned} \quad (17)$$

where $\mathbf{I}$ is the identity matrix, $\mathbf{U}_{ij}$ is defined in Table 1, $\Delta_{ij} = y_{ii}y_{jj} - y_{ij}^2 > 0$. By substituting $\theta = \delta(x_{ii}^{k+1})$ into (17), we can obtain:

$$\left[\mathbf{Z}(x_{ij}^{k+1})^{-1} \right]_{ij} = \frac{-\Delta_{ij} \delta(x_{ii}^{k+1}) + y_{ij}^k}{-\Delta_{ij} \delta(x_{ii}^{k+1})^2 + 2y_{ij}^k \delta(x_{ii}^{k+1}) + 1}. \quad (18)$$

When $s_{ij} = 0$, we substitute (18) into (16) and solving for x , the minimizer will be given by

$$m_{ij}^k = x_{ij}^k + \frac{y_{ij}^k}{\Delta_{ij}}. \quad (19)$$

When $s_{ij} \neq 0$, the minimizer will be given by:

$$\hat{x}_{ii}^{k+1} = m_{ij}^k = x_{ij}^k + \frac{y_{ij}^k}{\Delta_{ij}} + \frac{\Delta_{ij} - \sqrt{(\Delta_{ij}^k)^2 + 4s_{ij}^2 y_{ii}^k y_{jj}^k}}{2\Delta_{ij} s_{ij}}. \quad (20)$$

Therefore, if the two discontinues points are not in the domain of $\Psi(\cdot)$, i.e. $\det \mathbf{Z}(0) \leq 0$ and $\det \mathbf{Z}(x_{ij,t-1}) \leq 0$, we can have the expression of the minimizer from (19) and (20) as

$$\hat{x}_{ii}^{k+1} = m_{ij}^k = \begin{cases} x_{ij}^k + \frac{y_{ij}^k}{\Delta_{ij}} & \text{if } s_{ij} = 0 \\ x_{ij}^k + \frac{y_{ij}^k}{\Delta_{ij}} + \frac{\Delta_{ij} - \sqrt{(\Delta_{ij}^k)^2 + 4s_{ij}^2 y_{ii}^k y_{jj}^k}}{2\Delta_{ij} s_{ij}} & \text{if } s_{ij} \neq 0 \end{cases}. \quad (21)$$

Furthermore, we shall discuss the two continuous points in two different conditions respectively: $x_{ij,t-1} = 0$ and $x_{ij,t-1} \neq 0$.

Condition (1): if $x_{ij,t-1} = 0$

In this condition, the objective function can be expressed as

$$\begin{aligned} \Psi(x_{ij}^{k+1}) &= c(x_{ij}^{k+1}) \parallel \{x_{ij}^{k+1} \neq 0\} \\ &\quad + (c(x_{ij}^{k+1}) - 2(\lambda_1 + \lambda_2)) \parallel \{x_{ij}^{k+1} = 0\}. \end{aligned} \quad (22)$$

Thus, the minimizers are the minimizers of $c(\cdot)$ or just $x_{ij}^{k+1} = 0$. For notational simplicity, let $A = \Psi(0)$, $B = \Psi(x_{ij,t-1})$ and $C = \Psi(m_{ij}^k)$. Since the minimizer of $c(\cdot)$ is given above, we can summarize the solution for this condition as following:

- when $\det \mathbf{Z}(0) \leq 0$: the solution can be obtained from (21);
- when $\det \mathbf{Z}(0) > 0$:

$$\hat{x}_{ii}^{k+1} = \begin{cases} 0 & \text{if } A < C \\ m_{ij}^k \parallel \{\hat{x}_{ij}^k \neq 0\} & \text{if } A = C \\ m_{ij}^k & \text{if } A > C \end{cases}. \quad (23)$$

Condition (2): if $x_{ij,t-1} \neq 0$

Different from the previous condition, the objective function in this condition can be expressed as

$$\begin{aligned} \Psi(x_{ij}^{k+1}) &= c(x_{ij}^{k+1}) \parallel \{x_{ij}^{k+1} \neq 0\} \parallel \{x_{ij}^{k+1} \neq x_{ij,t-1}\} \\ &\quad + (c(x_{ij}^{k+1}) - 2\lambda_1) \parallel \{x_{ij}^{k+1} = 0\} \\ &\quad + (c(x_{ij}^{k+1}) - 2\lambda_2) \parallel \{x_{ij}^{k+1} = x_{ij,t-1}\}. \end{aligned} \quad (24)$$

In this case, the minimizers of the objective function are the minimizers of $c(\cdot)$ or $x_{ij}^{k+1} = 0$ or $x_{ij}^{k+1} = x_{ij,t-1}$. We can summarize the solution as following:

- when $\det \mathbf{Z}(0) \leq 0$ and $\det \mathbf{Z}(x_{ij,t-1}) \leq 0$: the solution can be obtained from (21);
- when $\det \mathbf{Z}(0) > 0$ and $\det \mathbf{Z}(x_{ij,t-1}) \leq 0$: the solution can be obtained from (23);
- when $\det \mathbf{Z}(0) \leq 0$ and $\det \mathbf{Z}(x_{ij,t-1}) > 0$:

$$\hat{\mathbf{x}}_{ii}^{k+1} = \begin{cases} \mathbf{x}_{ij,t-1} & \text{if } B < C \\ m_{ij}^k \parallel \{\hat{\mathbf{x}}_{ij}^k \neq \mathbf{x}_{ij,t-1}\} & \text{if } B = C \\ m_{ij}^k & \text{if } B > C \end{cases} \quad (25)$$

- when $\det \mathbf{Z}(0) > 0$ and $\det \mathbf{Z}(\mathbf{x}_{ij,t-1}) > 0$:

$$\hat{\mathbf{x}}_{ii}^{k+1} = \begin{cases} 0 & \text{if } A < B \text{ and } A < C \\ \mathbf{x}_{ij,t-1} & \text{if } B < A \text{ and } B < C \\ m_{ij}^k & \text{if } C < A \text{ and } C < B \\ \mathbf{x}_{ij,t-1} \parallel \{\hat{\mathbf{x}}_{ij}^k \neq 0\} & \text{if } A = B < C \\ m_{ij}^k \parallel \{\hat{\mathbf{x}}_{ij}^k \neq \mathbf{x}_{ij,t-1}\} & \text{if } A = C < B \\ m_{ij}^k \parallel \{\hat{\mathbf{x}}_{ij}^k \neq 0\} & \text{if } B = C < A \\ m_{ij}^k \parallel \{\hat{\mathbf{x}}_{ij}^k \neq \mathbf{x}_{ij,t-1}\} \parallel \{\hat{\mathbf{x}}_{ij}^k \neq 0\} + \mathbf{x}_{ij,t-1} \parallel \{\hat{\mathbf{x}}_{ij}^k = \mathbf{x}_{ij,t-1}\} & \text{if } A = B = C \end{cases} \quad (26)$$

In (25) and (26), A and B can be expressed as:

$$A = \Psi(0) = -\log \det(\mathbf{Z}(0)) + 2\lambda_2, \quad (27)$$

$$B = \Psi(\mathbf{x}_{ij,t-1}) = -\log \det(\mathbf{Z}(\mathbf{x}_{ij,t-1})) + 2s_{ij}\mathbf{x}_{ij,t-1} + 2\lambda_1. \quad (28)$$

2.2.4. Update $\mathbf{Y}^k$

When calculating the updated entries of $\mathbf{X}$, the inverse $\mathbf{Y}$ also needs to be updated as well in (11) and (21). Let $\theta = \delta(\mathbf{x}_{ij}^{k+1}) = \mathbf{x}_{ij}^{k+1} - \mathbf{x}_{ij}^k$ and $\Delta_{ij}^k = \mathbf{y}_{ij}^k \mathbf{y}_{ij}^k - (\mathbf{y}_{ij}^k)^2$, according to Sherman-Morrison-Woodbury formula and by assuming that $-\Delta_{ij}^k \theta^2 + 2\mathbf{y}_{ij}^k \theta + 1 \neq 0$, we have:

$$\mathbf{Y}^{k+1} = \begin{cases} \mathbf{Y}^k - \frac{\partial \mathbf{y}_{ij}^k (\mathbf{y}_{ij}^k)^T}{1 + \partial \mathbf{y}_{ij}^k} & \text{if } i = j \\ \mathbf{Y}^k - \frac{\theta \begin{bmatrix} \mathbf{y}_{ij}^k & \mathbf{y}_{ij}^k \end{bmatrix} \begin{bmatrix} 1 + \partial \mathbf{y}_{ij}^k & -\partial \mathbf{y}_{ij}^k \\ -\partial \mathbf{y}_{ij}^k & 1 + \partial \mathbf{y}_{ij}^k \end{bmatrix} \begin{bmatrix} (\mathbf{y}_{ij}^k)^T \\ (\mathbf{y}_{ij}^k)^T \end{bmatrix}}{-\Delta_{ij}^k \theta^2 + 2\mathbf{y}_{ij}^k \theta + 1} & \text{if } i \neq j \end{cases} \quad (29)$$

The convergence analysis of L0-SSR method is similar to the procedures that are provided in [38]. Interested readers are referred to [38] for more details.

2.3. Selection of regularization parameters λ_1 and λ_2

As mentioned in the previous sections, the regularization parameters λ_1 and λ_2 respectively control the degrees of spatial sparsity and temporal smoothness of the time-varying graphical Gaussian model. Meanwhile, the regularization terms also help to reduce the variance of the estimation so that the overfitting problem can be avoided. Due to these reasons, the selection of the regularization parameters greatly affects the estimate results. In previous related studies, the traditional parameter selection methods, such as Akaike information criterion (AIC) and Bayesian information criterion (BIC), were adopted [2,43,55,47]. However, these methods have their drawbacks. For example, AIC is susceptible to overfitting and BIC is valid only when the sample size is larger than the dimension of the estimated parameters [17]. In a high dimensional problem with relatively low sample size like this study, these methods are not ideal.

Table 2

The proposed L0-SSR method for dynamic inverse covariance estimation.

Step	Operation
1	Calculate the dynamic sample covariance matrices $\mathbf{S}_t$ using the sliding window-based method
2	Initialization. At $t = 1$, calculate $\mathbf{X}_1$ using the static inverse covariance estimation method proposed in [38]. At each timepoint $t = 2, \dots, T$, set the iteration number k to 1 and initialize $\mathbf{X}_t^k$ as $\text{diag}(1/s_{11,t}, 1/s_{22,t}, \dots, 1/s_{nn,t})$.
3	At each k -th iteration, $i, j = 1, 2, \dots, n$, update all entries of the inverse covariance matrix $\mathbf{X}$ as: 1) If $i = j$, calculate $\det \mathbf{Z}(\mathbf{x}_{ij,t-1})$. • If $\det \mathbf{Z}(\mathbf{x}_{ij,t-1}) \leq 0$, update $\hat{\mathbf{x}}_{ij}^{k+1}$ as (12). • If $\det \mathbf{Z}(\mathbf{x}_{ij,t-1}) > 0$, update $\hat{\mathbf{x}}_{ij}^{k+1}$ as (13). 2) If $i \neq j$, calculate $\det \mathbf{Z}(0)$ and $\det \mathbf{Z}(\mathbf{x}_{ij,t-1})$ respectively. • If $\mathbf{x}_{ij,t-1} = 0$ and $\det \mathbf{Z}(0) \leq 0$, update $\hat{\mathbf{x}}_{ij}^{k+1}$ as (21). • If $\mathbf{x}_{ij,t-1} = 0$ and $\det \mathbf{Z}(0) > 0$, update $\hat{\mathbf{x}}_{ij}^{k+1}$ as (23). • If $\mathbf{x}_{ij,t-1} \neq 0$, $\det \mathbf{Z}(0) \leq 0$, and $\det \mathbf{Z}(\mathbf{x}_{ij,t-1}) \leq 0$, update $\hat{\mathbf{x}}_{ij}^{k+1}$ as (21). • If $\mathbf{x}_{ij,t-1} \neq 0$, $\det \mathbf{Z}(0) > 0$, and $\det \mathbf{Z}(\mathbf{x}_{ij,t-1}) \leq 0$, update $\hat{\mathbf{x}}_{ij}^{k+1}$ as (23). • If $\mathbf{x}_{ij,t-1} \neq 0$, $\det \mathbf{Z}(0) \leq 0$, and $\det \mathbf{Z}(\mathbf{x}_{ij,t-1}) > 0$, update $\hat{\mathbf{x}}_{ij}^{k+1}$ as (25). • If $\mathbf{x}_{ij,t-1} \neq 0$, $\det \mathbf{Z}(0) > 0$, and $\det \mathbf{Z}(\mathbf{x}_{ij,t-1}) > 0$, update $\hat{\mathbf{x}}_{ij}^{k+1}$ as (26).
4	Update $\mathbf{Y}_t^k$ as (29).
5	If $\ \mathbf{X}_t^{k+1} - \mathbf{X}_t^k\ $ is less than a pre-set tolerance (usually set as a very small number, e.g. 10^{-6}), then go to Step 3 for the next timepoint, increment the timepoint counter t by 1. Otherwise, go to Step 4, increment the iteration counter k by 1.

In this study, we follow [17] to employ an extend Bayesian information criterion (EBIC) method for parameters selection. For any pair of λ_1 and λ_2 , we define the EBIC as:

$$\epsilon(\lambda_1, \lambda_2) = \sum_{t=1}^T \left(-\log \det(\mathbf{X}_t) + \text{tr}(\mathbf{S}_t \mathbf{X}_t) + \frac{\log p}{p} K(t) + 4\alpha \frac{\log n}{p} K(t) \right), \quad (30)$$

where $\alpha \in [0, 1]$ and is typically chosen as 0.5, p is the number of sample size, n is the dimension of the inverse covariance matrix, $K(t)$ is defined as the number of non-zero entries to be estimated at timepoint t . $K(t)$ can be expressed as:

$$K(t) = \begin{cases} \sum_{i \leq j} \parallel \{\mathbf{x}_{ij,t} \neq 0\} & \text{if } t = 1 \\ \sum_{i \leq j} \parallel \{\mathbf{x}_{ij,t} \neq 0\} \cap \parallel \{\mathbf{x}_{ij,t-1} \neq 0\} & \text{if } t \neq 1 \end{cases} \quad (31)$$

After the evaluation of all pairs of λ_1 and λ_2 , the preferred pair is the one with the minimum EBIC $\epsilon(\lambda_1, \lambda_2)$.

2.4. Summary of the CD-based algorithm for L0-SSR method

Here we summarize the proposed method for minimizing (2) in Table 2 for a given pair of λ_1 and λ_2 .

3. Simulation results

3.1. Simulation settings

In this section, the performance of the L0-SSR method is evaluated and compared to well-established dFC estimation methods on synthetic multivariate time-series data, which are generated based on different network topology properties [45].

To simulate various types of FC structures and time-varying properties, we conduct three multivariate simulations using different types of graph topologies: a small-world graph from the Watts-Strogatz (WS) model [56], a scale-free graph from the Barabasi-Albert (BA) model [5] and a random graph from the

Erdos-Renyi (ER) model [24]. These three models are chosen because that small-world and scale-free are two commonly seen graphs in real FCN [14,8,9,23,11] while ER random networks are the most widely used random networks [40].

In order to comprehensively validate the performance of our method, we conduct the simulation experiments with two types of network sizes: a small-size network with 10 nodes and a relatively large network with 50 nodes. For each simulation, we generate multivariate time-series data that consists of 3 segments, and each segment has 100 timepoints (in total 300 timepoints). During each segment, the network topology is randomly generated using one graph model (WS, BA, or ER). For the WS small-world graphs, the parameter that quantifies the probability of an edge rewiring randomness is set to 0.5, and the mean node degree is set to 2. For the BA scale-free graphs, the network generation begins with an initial seed network of size 5, and the remaining nodes are introduced sequentially into the network using preferential attachment model. For the ER random graphs, the probability of an edge between two nodes is fixed at 0.2. In the end, each time-series signal contains 2 change points at timepoint 100 and 200, and the network topology remains constant within each segment. The primary interest of this simulation is to examine whether the proposed method can correctly retrieve the topology that underlies the synthetic BOLD signals.

For performance evaluation, we first compare the dFC estimates to the ground-truth as a binary classification test at each timepoint using the following metrics: true positive (TP, correctly estimated connections), true negative (TN, correctly estimated no-connections), false positive (FP, incorrectly estimated connections) and false negative (FN, incorrectly estimated no-connections). Based on these four metrics, we can further measure the precision and recall of the estimation, which are defined as

$$\text{Precision} = \frac{TP}{TP + FP}, \quad (32)$$

$$\text{Recall} = \frac{TP}{TP + FN}. \quad (33)$$

Precision and recall indicate the correctly identified connections among all estimates and among all true connectivity, respectively. Ideally, we would like to have both recall and precision equal to 1. Furthermore, we can combine precision and recall together in the F1-score to provide a single and general measurement for imbalanced classes (i.e., the numbers of connections and no-connections are largely different). F1-score is calculated as:

$$\text{F1-score} = 2 \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (34)$$

Secondly, we also include the mean squared error (MSE) to measure the bias between the ground-truth of FC levels and the estimates from different methods. Due to the symmetrical property of the FC matrices, we only consider the upper triangle of each FC matrix for the identified FCs. The MSE can be calculated as:

$$\text{MSE} = \frac{1}{n(n-1)/2} \sum_{i=1}^n \sum_{j=i+1}^n \left([\mathbf{X}_t]_{ij} - [\hat{\mathbf{X}}_t]_{ij} \right)^2, \quad (35)$$

where $\mathbf{X}_t$ is the ground-truth FC matrix at timepoint t , $\hat{\mathbf{X}}_t$ is the corresponding identified estimates, and n is the dimension of the FC matrix. Since the MSE directly measures the FC levels, it can describe the distance from the estimates to the ground-truth. MSE is useful because it is sometimes more interesting to investigate FC's level instead of determining its existence only.

The proposed L0-SSR method is compared to two conventional sparsity-promoting dFC estimation methods, namely Smooth Incremental Graphical Lasso Estimation (SINGLE) [42,41] and slid-

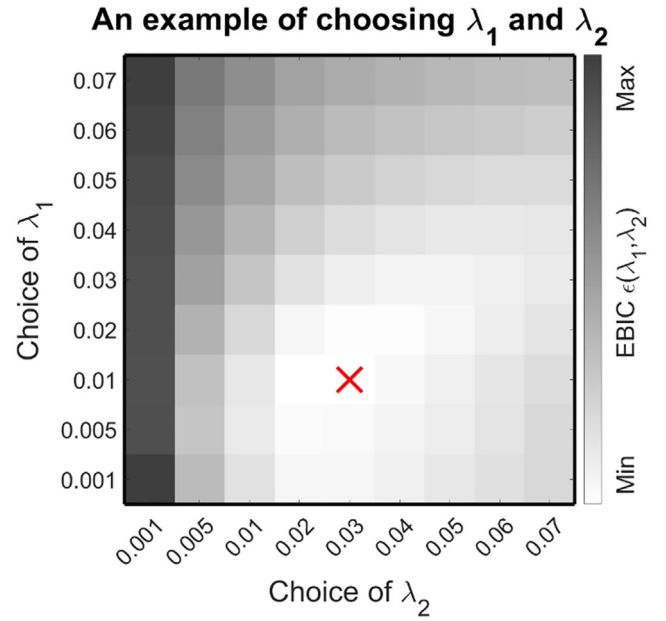


Fig. 1. An example to demonstrate the selection of regularization parameters. EBIC calculated from different choices of λ_1 and λ_2 is color coded. The block with a red cross is the minimal point of EBIC, where the corresponding λ_1 and λ_2 are finally chosen as the regularization parameters.

ing window-based Time-Varying Graphical Lasso (TVGL) [25,15]. These two methods are chosen because they are both based on sparse inverse covariance estimation with the Gaussian graphical model and use ℓ_1 -norm to obtain sparse estimation. All simulation procedures are repeated for 100 times to obtain the mean and standard error of mean (SEM) for comparison.

3.2. Selection of regularization parameters using EBIC

The regularization parameters λ_1 and λ_2 control the weight imposed into the original model for spatial sparsity and temporal smoothness constraints respectively. As stated in Section 2.3, these parameters are determined by using the EBIC method [17]. Fig. 1 shows an example result of choosing λ_1 and λ_2 . In the figure, EBIC $\epsilon(\lambda_1, \lambda_2)$ are calculated from equation (30) for different combinations of regularization parameters, and are color coded. The selected pair of λ_1 and λ_2 that provides the minimal EBIC is indicated by the red cross. This typical example is chosen from one simulation of WS small-world network for a brief demonstration. It should be noted that the choice of regularization parameters would be different for different simulations.

3.3. Results

Figs. 2–4 show the multivariate simulation results from small-world, scale-free and random networks, respectively. The averaged F1-scores and MSE from 100 realizations are used to evaluate the performances of different methods, which are respectively shown in the upper and lower panels of Figs. 2–4. Further, we compare the simulation results with 10 nodes and 50 nodes for each type of network and show the results of 10 and 50 nodes respectively in left and right panels of Figs. 2–4. To make a closer observation of the FC estimates of the methods under comparison, we selected and illustrated several representative examples of dFC estimates from each simulation setting in Fig. 5.

From the F1-scores and MSE results, we can see that the proposed L0-SSR method performs the best among three methods

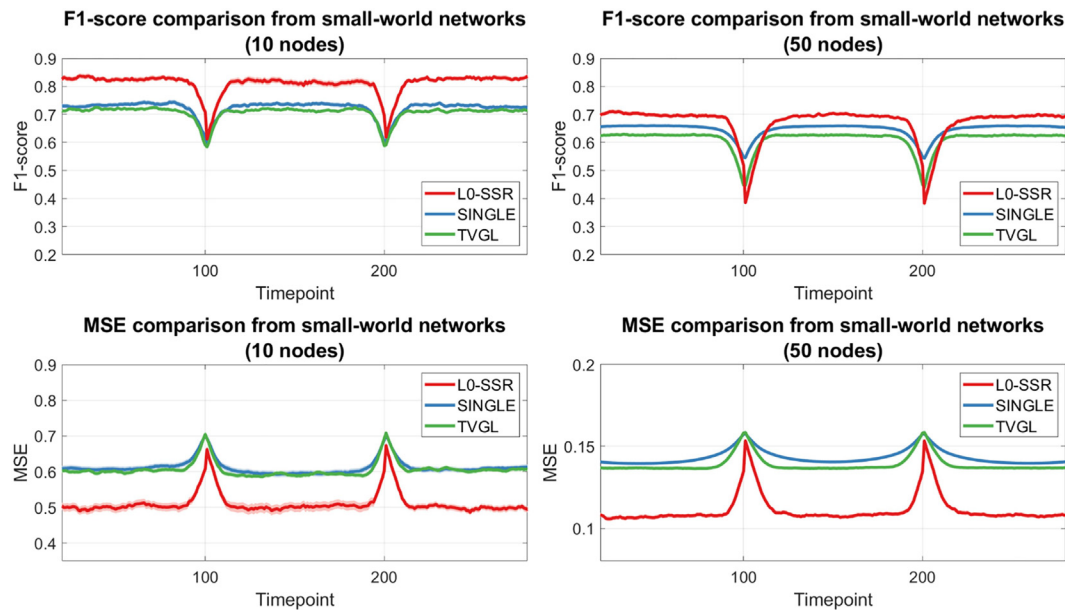


Fig. 2. Comparison of F1-scores and MSE among three methods on simulated small-world networks with different numbers of nodes (10 and 50). Mean and SEM of different methods are shown as lines and shades in different colors. The network structures have two sudden changes at timepoints 100 and 200.

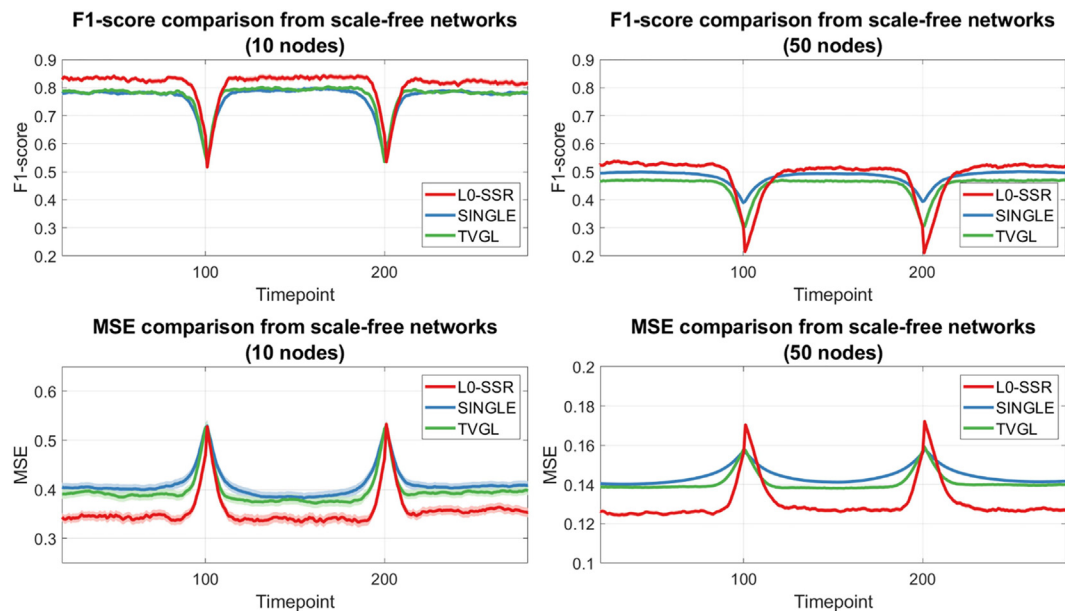


Fig. 3. Comparison of F1-scores and MSE among three methods on simulated scale-free networks with different numbers of nodes (10 and 50). Mean and SEM of different methods are shown as lines and shades in different colors. The network structures have two sudden changes at timepoints 100 and 200.

under comparison from small-world and scale-free networks shown in both Figs. 2 and 3. In random networks, the proposed method achieves a comparable performance to SINGLE and TVGL in terms of F1-scores from 10 nodes networks, but is still better than them for MSE comparison. Although three methods can all track the network changes soon after the transition occurrence, the proposed L0-SSR method is largely unaffected and can keep its performance at a relatively high level longer than other two methods. Meanwhile, L0-SSR method can retrieve accurate estimates after the network structure changes more quickly than other methods. These results demonstrate the effectiveness of the proposed L0-SSR method in identifying time-varying FC without the need of manually splitting the data.

In the Figs. 2–4, we note that performance degrades at timepoints 100 and 200, respectively, because the network structure has suddenly changed at these two timepoints. This can be clearly seen from Fig. 5. More specifically, the synthetic signal (shown in black lines) changed its magnitude at timepoints 100 and 200. Thus, the entire network topology is changed from one form to another correspondingly. The reason of this simulation design is to simulate the dynamic changes of brain network in different conditions. The proposed method as well as other two methods under comparison all use prior information from neighboring timepoints estimation. When dealing with the problem of topology sudden change, the regularization may introduce inaccurate prior information for current estimation which could affect the prediction accu-

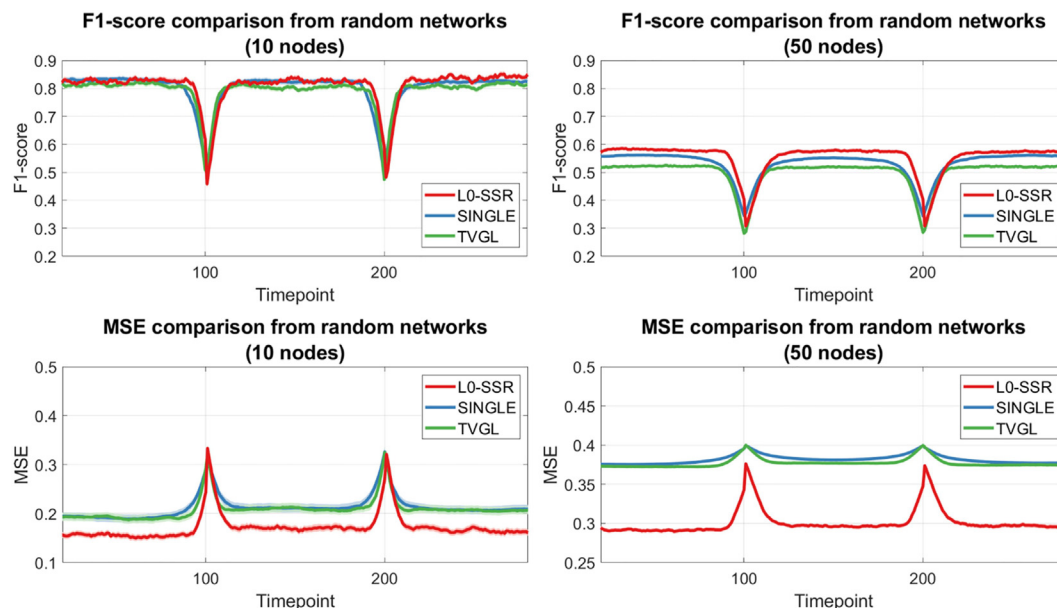


Fig. 4. Comparison of F1-scores and MSE among three methods on simulated random networks with different numbers of nodes (10 and 50). Mean and SEM of different methods are shown as lines and shades in different colors. The network structures have two sudden changes at timepoints 100 and 200.

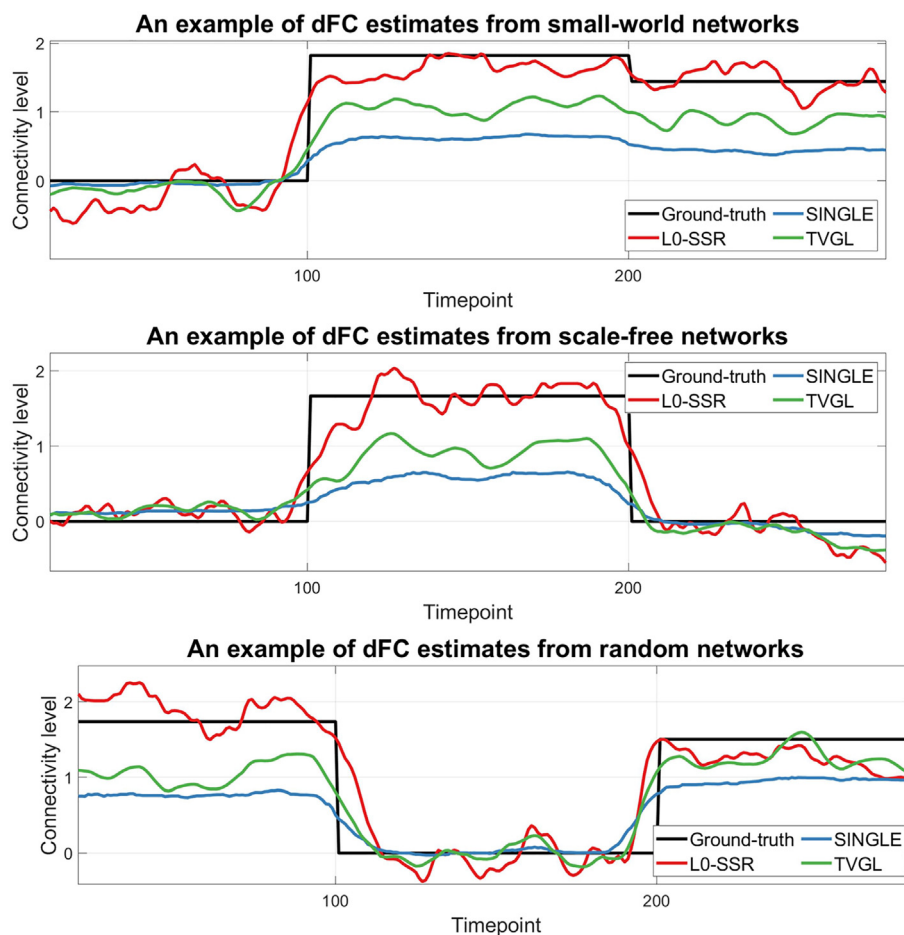


Fig. 5. Three examples of dFC estimation from simulations of small-world, scale-free and random networks. The estimated connectivity levels of different methods are shown in different colors. The ground-truth is shown in black.

racy largely. As so, the performance will drop around the change point. However, a good method (such as the proposed method) should be able to adjust the estimation rapidly. From Figs. 2–4 of the manuscript, we can see that our proposed method can track back to the true FC faster than other two competitors after the occurrences of FC changes. Meanwhile, as observe from these examples, the proposed method can track the dFC with higher accuracy. Though the other two methods can both follow the trend of dFC, the results from the proposed method is closer to the ground-truth.

It should be noted that the SINGLE method gives a much smoother result than LO-SSR and TVGL, which is probably because it prefers to choose a larger window size for estimation. More specifically, SINGLE is an extension of a sliding window method with regularization terms, and it can automatically select its window length by using a cross-validation method [40]. However, the CV selection method tends to obtain a relatively large window size so that the estimation can be more robust for the entire time-series. But, the drawback of such a choice is that the ability of tracking instantaneous changes is degraded, as shown in the blue lines in Fig. 5.

4. Application to motor-task fMRI dataset

4.1. Data acquisition and preprocessing

In this section, we apply the proposed LO-SSR method to a motor task fMRI dataset. The dataset is obtained from the Human Connectome Project (HCP). For more details about HCP data, please refer to the HCP website: <https://db.humanconnectome.org/>. The entire dataset can be downloaded from <https://db.humanconnectome.org/app/action/DownloadPackagesAction>. The experiment was accomplished using a block-design paradigm, in which 45 participants were asked to perform a series of motor tasks [6,13,60]. More specifically, the participants were asked to conduct five different simple motor tasks including: tap their left fingers (LF) or right fingers (RF), squeeze their left toes (LT) or right toes (RT), or move their tongues (TG), after a visual cue word was presented on the screen at the beginning of each block. The paradigm contained 2 runs with 13 blocks each: 10 of them were motor tasks (each having a length of 15 s, including a 3 s cue and a 12 s movement), and the other 3 were fixation (FX) blocks. The order of the blocks in one run was: 1. RF, 2. LT, 3. TG, 4. RT, 5. LF, 6. FX, 7. TG, 8. LT, 9. RF, 10. FX, 11. LF, 12. RT, 13. FX. The data were acquired with a 3 T Siemens Skyra with TR = 720 ms.

The preprocessing procedures are carried out using SPM12 (<https://www.fil.ion.ucl.ac.uk/spm>) and DPABI (<http://rfmri.org/dpabi>) [58] in MATLAB 2017a environment. Before preprocessing, first 10 volumes are removed to avoid T1 equilibration effect. The remaining images are processed by slice timing correction and realignment for head-motion correction. The functional

images are then co-registered to subjects' high-resolution anatomical images and normalized into a standard Montreal Neurological Institute (MNI) space. All functional images are smoothed using a Gaussian kernel (Full Width at Half Maximum FWHM = 8 mm). The linear trend and other nuisance signals (including Friston's 24 motion parameters, cerebrospinal fluid, white matter and global signals) are also regressed out [27]. Finally, a high-pass filter with cut-off frequency of 0.005 Hz is used to reduce the interference of low-frequency noise. In the end, in total 271 frames (around 195 s) are obtained for each subject.

4.2. Analysis methods

After the preprocessing, we extract the BOLD time-series from a set of selected ROIs for studying the dFC. We follow [41] to define 22 cortical ROIs (11 for each cerebral hemisphere) covering occipital, parietal and temporal lobes based on Desikan-Killiany atlas [21]. The names and coordinates for each ROI can be found in Table 3. We target these brain regions because their BOLD responses and FC changes are typically found in related to motor tasks [41]. By applying the proposed LO-SSR method on the dataset, we obtain a dynamic FC matrix of size $22 \times 22 \times 271$ for each subject. Due to the symmetrical property of FC, the subsequent analyses will only consider the upper triangle of the matrices at each timepoint. We carry out two analyses to examine the performances of the proposed method and other methods under comparison.

4.2.1. Comparison between dFC-based clustered states and task blocks

Firstly, we investigate whether the dynamic FCNs appear as clusters under different motor tasks. Clustering is the most popular way to identify the reoccurring brain network patterns (so called "brain states") within a certain task or rest condition. In our work, *k*-means clustering is applied to time-varying FCN estimates for capturing reoccurring brain states. Because the task periods are known, FCN from the same task were averaged as the initialized centroids for *k*-means. With the results of *k*-means clustering, we check whether the estimated FC networks can be correctly clustered into corresponding task conditions.

4.2.2. Identification of task-evoked FC of motor-related ROIs

Secondly, we investigate the characterized FC patterns under motor tasks. All FCNs estimated by LO-SSR can be assigned into 2 categories: motor tasks and fixation without any particular task. To characterize the task-evoked FCN patterns, we first average all FCN across all subjects to obtain relatively stable and more general estimation results. We consider fixation (FX) as a baseline since there are no tasks in FX blocks except spontaneous activity. Further, for each individual FC (i.e. each entry from the estimated matrices) from tasks, we compare the FC value from task blocks to the baseline to check whether they are task-related FC. Since the experiment has various motor tasks, we combine all different

Table 3

The names and MNI coordinates of selected 22 ROIs. Each ROI is numbered for convenient identification in the subsequent contents.

Name of ROI	No.	Coordinate (left)		No.	Coordinate (right)	
Lateral Occipital	1	−29	−87	1	31	−84
Inferior Parietal	3	−39	−68	30	4	43
Superior Parietal	5	−21	−64	47	6	22
Precuneus	7	−10	−57	37	8	11
Fusiform	9	−34	−43	−19	10	34
Lingual	11	−14	−67	−3	12	15
Inferior Temporal	13	−49	−31	−23	14	49
Middle Temporal	15	−56	−27	−12	16	57
Precentral	17	−38	−9	43	18	39
Postcentral	19	−42	−23	44	20	42
Paracentral	21	−8	−28	59	22	9
						−26
						58

tasks together and compare the whole task period to the baseline. Also, because of all tasks in the experiment are motor tasks, we focus on FC between the motor-related ROIs (such as precentral regions) and other regions. We apply a two-tailed t -test to examine whether the FC level during task stimulation is different from fixation.

4.3. Experimental results

4.3.1. Comparison between dFC-based clustered states and task blocks

In Fig. 6(a), the clustering results from different methods are shown together with the experimental task design (i.e., the ground truth). We can briefly see that the proposed L0-SSR method has the highest accuracy from clustering where its FCN estimates can generally distinguish different task types. More specifically, the clustering accuracy for L0-SSR can reach 84.87%, while the other two methods only have 54.61% and 80.07% for SINGLE and TVGL respectively. Moreover, the clustering confusion matrices from three methods are shown in Fig. 6(b). An ideal clustering would have a confusion matrix with non-zeros shown in red blocks on the diagonal entries (correct clustering) and zeros shown in white blocks in the off-diagonal entries (errors). From Fig. 6 we can conclude that the L0-SSR gives the smallest errors among three methods. These results illustrate that the L0-SSR method can accurately capture the brain network variations across different tasks, indicating that it is a reliable and powerful approach to characterize dynamic FC from real fMRI data.

4.3.2. Task-evoked FC of motor-related ROIs

Fig. 7(a) shows a set of dynamic FCs from L0-SSR that are connected with motor-related regions. To make a better presentation, we smooth the dFC using a rectangular window of length 30TR. Moreover, Fig. 7(b) shows the statistical comparison of dFC levels between tasks and baseline for those dynamic FCs shown in Fig. 7(a) respectively. The t -test results are adjusted according to false discovery rate (FDR) using Benjamini-Hochberg procedure at $\alpha = 0.001$ level. Similar results from SINGLE and TVGL can be found in Fig. 8. It is interesting to observe that each dFC increases to its maxima within the task blocks, while decreases to its minimum within fixation blocks. This is particularly obvious for the FC between region 18 (right precentral) and region 1 (left lateral occipital), as well as the FC between region 18 (right precentral) and region 5 (left superior parietal).

5. Discussion and conclusion

5.1. Contribution of this work

In this work, we introduce a novel dynamic FC estimation method which is based on inverse covariance estimation with newly proposed ℓ_0 -norm-based spatial sparsity and temporal smoothness regularizations. The superior performance of the proposed method is evaluated using synthetic time-series data and a real fMRI dataset from a block-design motor task experiment.

From the model of the proposed L0-SSR method as presented in Eq. (2), we can see that the proposed model consists of three

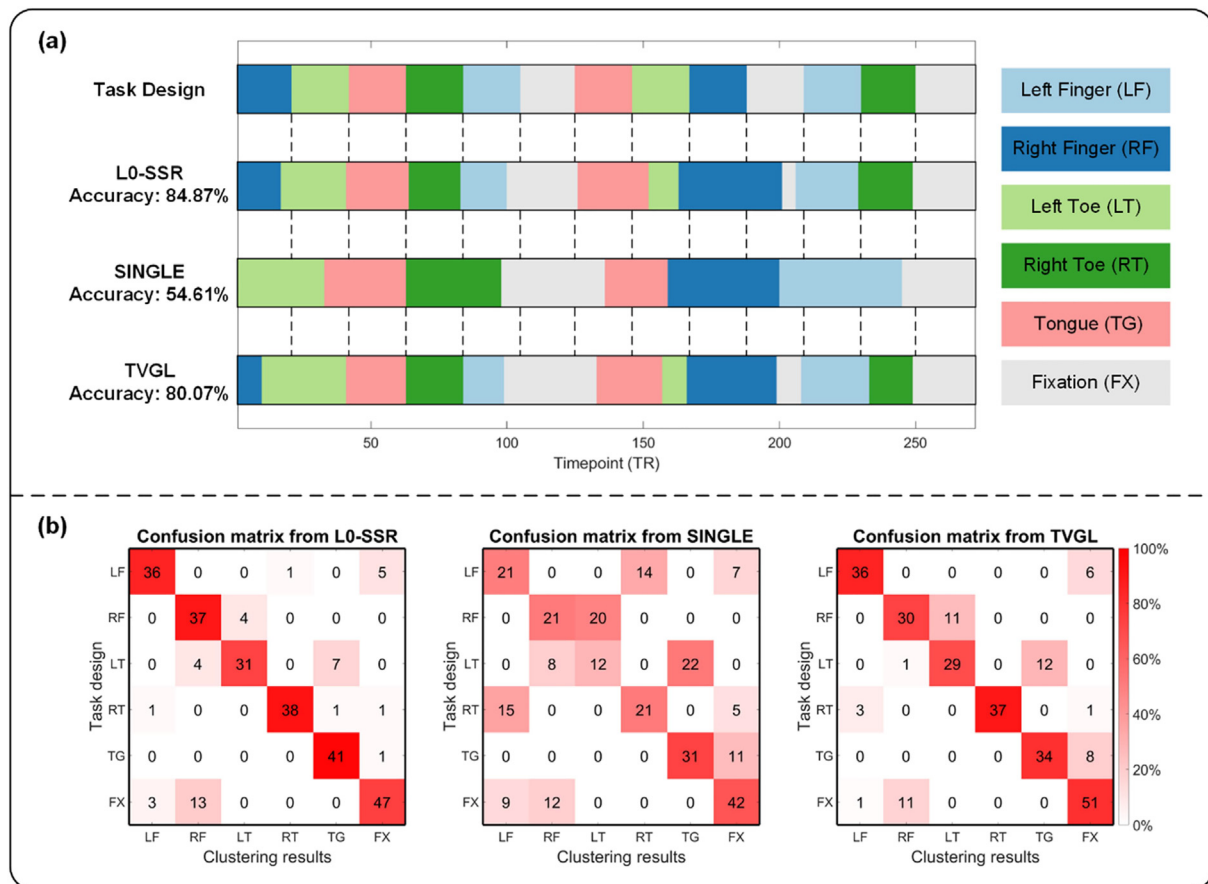


Fig. 6. (a) Clustering of dFC matrices estimated from three methods (L0-SSR, SINGLE and TVGL) and comparison with the task design (the ground-truth). (b) Confusion matrices of clustering results of three methods (L0-SSR, SINGLE and TVGL). Rows of each matrix correspond to different motor tasks and columns correspond to clustering results of different methods. The number in each block refers to the FC instances known to be in one task and clustered to be another task. The background of each block depicts the proportion of clustering (from 0 to 100%) for each pair of tasks.

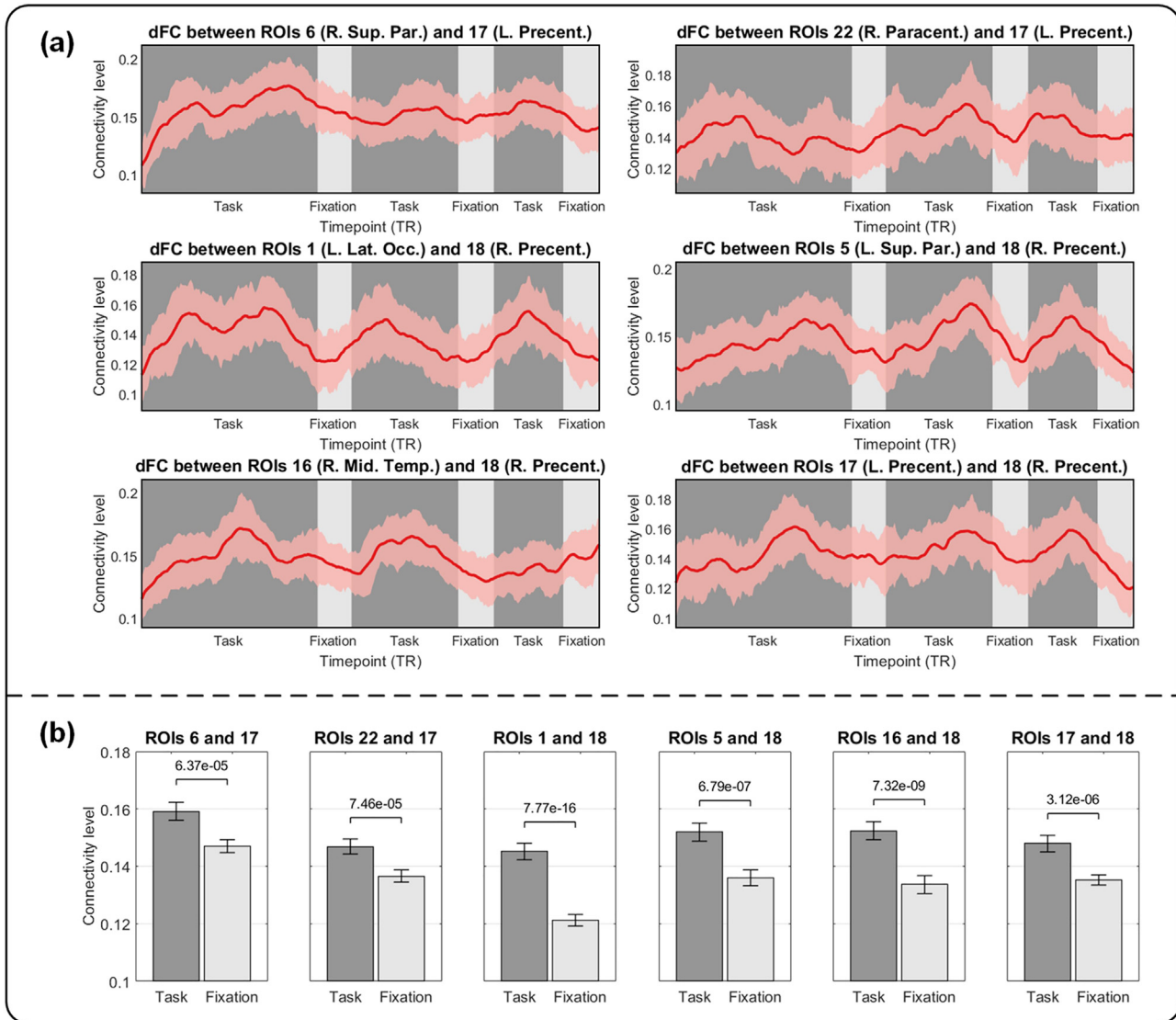


Fig. 7. (a) dFC of selected motor-related regions from L0-SSR. The darker grey background indicates the periods of movement tasks, while the lighter grey background indicates the fixation (baseline) blocks. The means and SEMs of FC are shown as red lines and shades. (b) Statistical comparison of motor-related dFC between tasks and baseline (fixation). Error bars stand for SEM. The p -value of each comparison is shown above corresponding bars.

parts: an ℓ_0 -norm-based spatial sparsity regularization term, an ℓ_0 -norm-based temporal smoothness regularization term and a time-varying dynamic Gaussian graphical model. The major advantage of the proposed L0-SSR method stems from the two ℓ_0 -norms used. Unlike the conventional ℓ_1 -norm regularization, the ℓ_0 -norm can enforce the strongest sparsity by shrinking the estimates to exact zero so that false positives can be effectively eliminated. Because only the most significant coefficients are remained under the ℓ_0 -norm, the proposed method can obtain sparser and more accurate estimates than ℓ_1 -norm regularization. On one hand, by using the ℓ_0 -norm-based temporal smoothness regularization, L0-SSR method can successfully enforce the similarity and continuity between two temporally-adjacent FC networks. On the other hand, by using the ℓ_0 -norm-based spatial sparsity regularization, L0-SSR method can effectively retain strong FC between brain regions while ignore those insignificant FC. Both ℓ_0 -norm based regularizations are inspired by biological reality, and thus the estimation result is similar to real-world brain networks. Third, many prior studies on dynamic FC inferring are closely related to the sliding window method [3,40,15],

one major limitation of this traditional but popular sliding window method is the uncertainty of results introduced by window length selection. Though the proposed L0-SSR method also uses sliding windows to get the sample covariance matrices, we avert the window selection problem in a manner of timepoint-by-timepoint estimation. This is largely contributed by the time-varying Gaussian graphical model which can separate each timepoint to a relatively minor estimation problem. Again, thanks to the temporal smoothness that introduces prior information for each timepoint estimation, such a scheme can also deal with overfitting well even when the number of samples is small and the dimension of estimation coefficients is high.

In order to solve the proposed model effectively, we further propose a computing framework based on the CD algorithm. The core idea of the CD-based algorithm is to iteratively solve the minimization problem for only one entry at each iteration while keep all other entries fixed. Unlike conventional ℓ_1 -norm regularization such as LASSO, our problem cannot be analytically solved. To address this, we discuss different conditions during the iterations and provide the minimizer for each condition. We also derive all

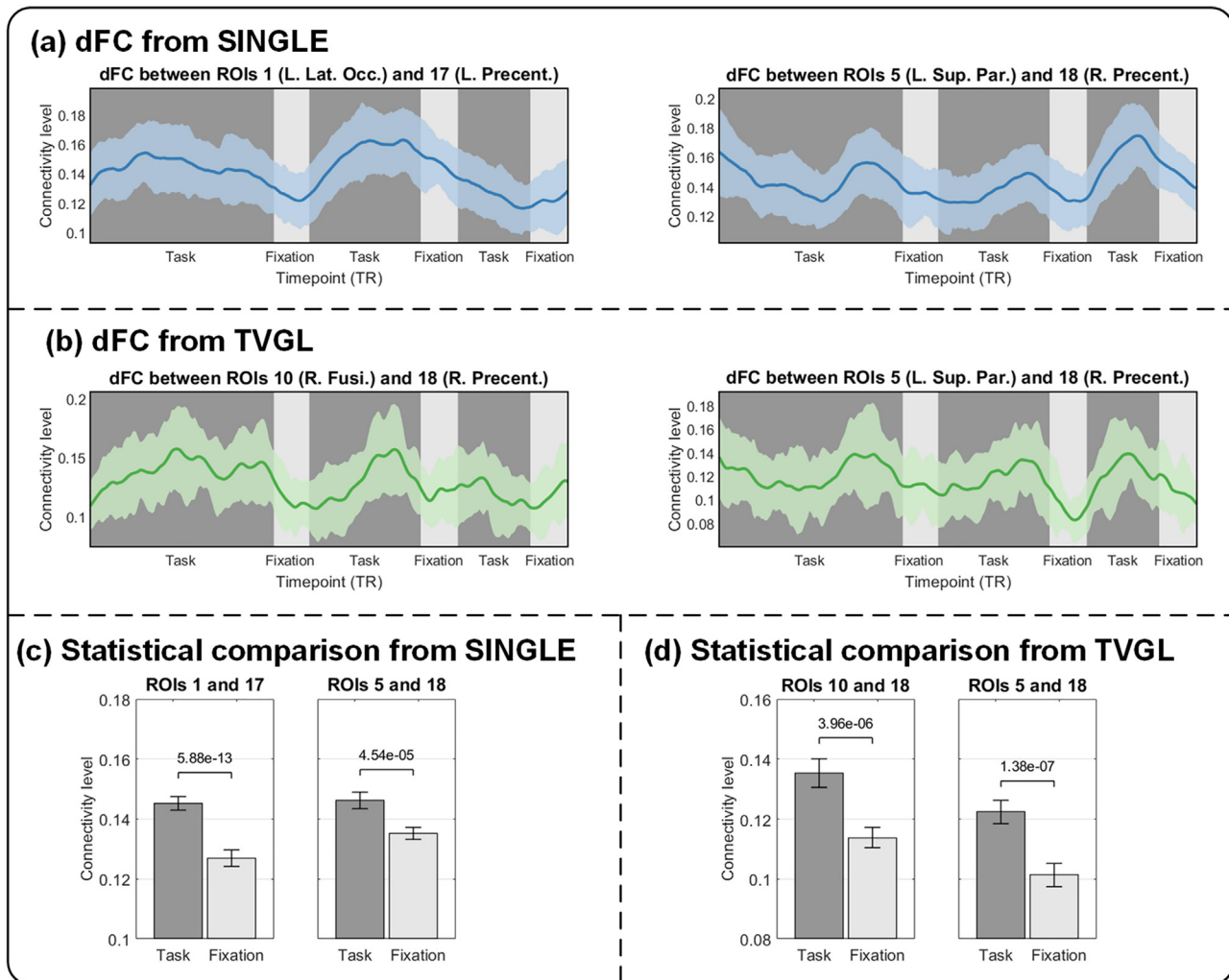


Fig. 8. (a) and (b): dFC of selected motor-related regions from SINGLE (blue) and TVGL (green) respectively. The darker grey background indicates the periods of movement tasks, while the lighter grey background indicates the fixation (baseline) blocks. The means and SEMs of FC are shown as lines and shades respectively. (c) and (d): statistical comparison of motor-related dFC between tasks and baseline (fixation) from SINGLE and TVGL respectively.

the steps mathematically in Section 2.2 and summarize the entire algorithm in Table 2.

5.2. Performance of L0-SSR on simulated and real data

To validate the effectiveness of the proposed L0-SSR model and the CD-based algorithm, we conduct a series of simulation studies in Section 3. More specifically, we use the WS model, the BA model and the ER model to generate small-world, scale-free and random networks, which resemble the different network configurations of real human brain. The human brain is a complex dynamic system with coevolutionary brain activity and connectivity and previous studies have shown small-world and scale-free organization of connectivity networks in the human brain [23,52,49]. These network simulation models allow us to synthesize fMRI time-series with different connectivity network properties so as to examine the performance of the proposed method from a more comprehensive perspective. Our simulation results show that the L0-SSR method can accurately estimate the underlying FCN structures from time-series data, which provides strong evidence for the superiority of the L0-SSR method over other sparsity-enhancing FC estimation methods.

More specifically, simulation results show that the proposed method has three advantages, which make it particularly suitable

for estimating dynamic FC of real-world fMRI. First, the L0-SSR method has smaller estimation bias than other methods under comparison. As shown in Fig. 5, the estimation curves of L0-SSR fluctuate around the ground-truth. Although the other two methods can follow the changing trends of the ground-truth, they both show relatively larger bias. The property of small biasedness of L0-SSR is very useful, because other than to examine the existence of FC, it is more meaningful to know the exact FC values, so that FC strengths among different ROIs can be compared. Second, our method is more sensitive to sudden changes of FC, and it can track back to the true FC levels more rapidly after the occurrences of FC changes. The excellent tracking performance of the proposed method is attributed to the time-varying Gaussian graphical model and the ℓ_0 -norm-based temporal smoothness regularization used. Above two advantages make L0-SSR particularly suitable for the FC estimation of task-related fMRI because tasks could produce remarkable changes of FC and it is necessary to compare FC strengths to find the strongest FC related to specific tasks. Third, the proposed L0-SSR method performs better than other well-established sparse dFC estimation methods, especially for the small-world and scale-free networks. This implies that the proposed method is suitable for real-world data because it has been widely reported that the FCN is not purely random but with small-world and/or scale-free properties [14,8,9,23,11].

The proposed method can also get more accurate and interpretable results on a public motor task fMRI dataset. Based on the estimated time-varying FCN for each timepoint, we investigate the FC patterns under different motor tasks. We can find that the FC concerning motor-related regions show meaningful patterns, suggesting possible underlying information transmission mechanism during motor tasks. By using the proposed method, the estimated dynamic FCNs can be more accurately clustered into different stimulation conditions. This finding suggests that our method can better capture the transient brain dynamics that are associated with different brain conditions. This might help to advance our understanding of neural mechanisms of different brain functions.

Regarding each FC pair, Figs. 7 and 8 show that some motor-related FC are significantly larger during tasks than baseline. There are many interesting phenomena can be found from these results. Firstly, by observing these dFC from different methods, we can find that there exist obviously larger FC values during the task blocks while the FC decrease during the baseline blocks for all significant dFC. This may indicate that there exists stronger information transfer among different brain regions during the motor-tasks. Secondly, by comparing the results obtained from three methods, we can see that all of them can find that the connectivity between ROIs 5 and 18 is significantly larger during tasks than fixation baseline. ROI 5 belongs to superior parietal lobe which is involved with visual input and sensory input, and ROI 18 belongs to precentral gyrus which is the site of primary motor cortex that deeply involved in human motor activities. This is interesting since it is a stable finding from different methods. One possible explanation of this finding is that during the task blocks, there exists stronger information transfer from occipital and superior parietal regions to primary motor cortex so that the participants can finish the required movement task according to the visual cue. Such an interpretation may demonstrate the involvement of high-level visual processing in motor functions as mentioned previously in [43,63]. Thirdly, aside from the dFC between motor-related ROIs and visual-related ROIs, the proposed method also suggests some meaningful information that may cannot be obtained using other two methods. A novel finding of our method is that there are two within-motor connectivities between ROIs 17 and 18 and between ROIs 17 and 22. These connectivities are linked among precentral and paracentral regions. It is known that the paracentral regions are involved with human motor and sensory innervations of the lower extremity. This may imply that, besides the information transferring increasing between motor-related regions to other outside regions, there exist stronger communications inside the motor-related regions. This phenomenon may refer to the self-related processing during motor tasks, which has an implication of the underlying neural mechanism of functional reorganization [32,44]. From the results presented by Figs. 7 and 8, we speculate that the motor task does not only activate these related regions, but also makes them more synchrony so that they can transfer motor information more effectively.

5.3. Limitations and future work

One major limitation of the proposed L0-SSR method is its large computational complexity, which cannot be realized in an online manner. In the present work, FC networks are estimated by optimizing the objective function with ℓ_0 -norm-based regularization terms. Though we have derived the algorithmic solution for various conditions during CD-based iterative steps, the entire calculation is still time consuming. One possible solution to lower the computational complexity of the proposed method is to disassemble the entire problem into smaller formulations for each timepoint. Similar idea can be found in [41], where a real-time dFC

inferring system was built by setting the network structure from previous timepoint as fixed. Meanwhile, one can find that the proposed model is a combination of the original Gaussian graphical model and two regularization terms. In such a circumstance, we can even tackle the regularized problem using augmented Lagrangian multiplier method which allows the penalties to be decoupled from the original inverse covariance estimation model. This can greatly simplify the problem and improve the computational efficiency by using more advanced distributed implementation, such as Split Bregman [59] and Alternating Direction Method of Multipliers [10]. By doing so, it is possible to achieve online realization of the proposed method, which holds a great potential for clinical uses, such as real-time fMRI FC neurofeedback.

5.4. Conclusion

In conclusion, a new L0-SSR method, which incorporates spatial sparsity and temporal smoothness ℓ_0 -norm-based regularization terms into the conventional inverse covariance estimation model, is developed for estimating dFC from fMRI time-series. Instead of using the most conventional ℓ_1 -norm, this work uses ℓ_0 -norm-based regularization terms to impose spatial sparsity and temporal smoothness into the estimation of dFC. The resultant optimization problem is solved using self-derived iterative steps based on the CD method. The proposed method is then evaluated using various synthetic fMRI data with different network properties and using real fMRI data recorded in motor tasks. Results from both simulation and real-world data suggest that the proposed method is a valuable tool for estimating dynamic FC networks.

CRediT authorship contribution statement

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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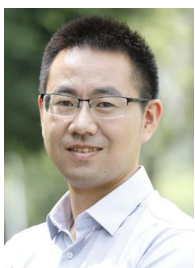
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